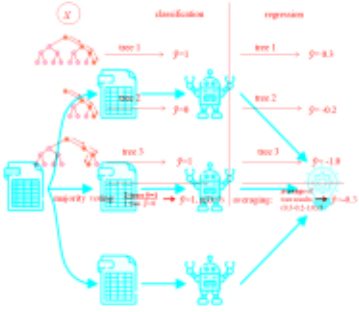
Introduction to Machine Learning

Random Forests: Bagging Ensembles

Bagging Ensembles



Learning goals

- Understand the basic idea of bagging
- Be able to explain the connection of bagging and bootstrap
- Understand how a prediction is computed for bagging
- Understand why bagging improves the predictive power

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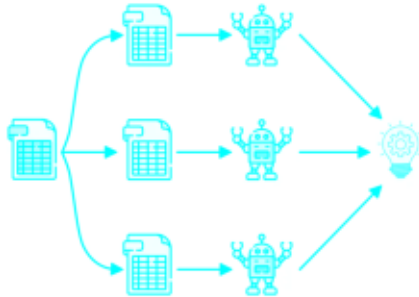
- Understand why bagging improves predictive performance

BAGGING

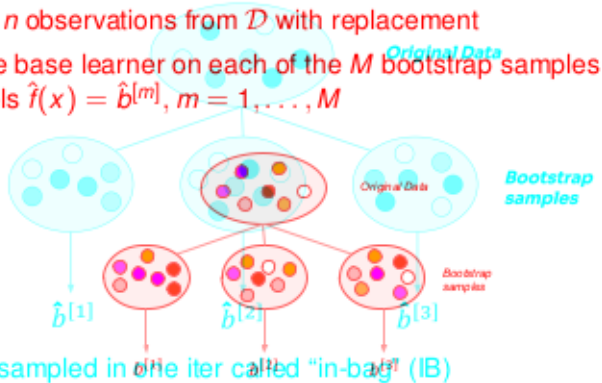
- Bagging is short for **B**ootstrap **Agg**regation.
- It's an **ensemble method**, i.e., it combines many models into one big "meta-model"
- Such model ensembles often work much better than their members alone would.
- The constituent models of an ensemble are called **base learners**



BAGGING / BAGGED ENSEMBLES

In a **bagging** ensemble all base learners are of the same type. The only difference between the models is the data they are trained on. Specifically, we train base learners $b^{[m]}$, $m = 1, \ldots, M$ on M **bootstrap** samples of training data $\mathcal{D}$:

- Draw n observations from $\mathcal{D}$ with replacement
- Fit BL on each bootstrapped data $\mathcal{D}^{[m]}$ to obtain $b^{[m]}$
- Draw n observations from $\mathcal{D}$ with replacement
- Fit the base learner on each of the M bootstrap samples to get models $\hat{f}(x) = \hat{b}^{[m]}, m = 1, \ldots, M$

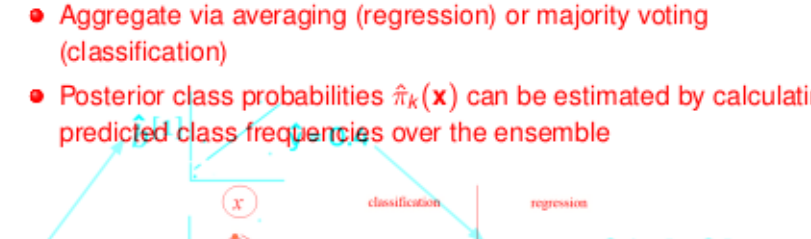


- Data sampled in one iter called "in-bag" (IB)
- Data not sampled called "out-of-bag" (OOB)

BAGGING WITH A BAGGED ENSEMBLE

Aggregate the predictions of the M fitted base learners to get the **ensemble model** $\hat{f}^{[M]}(\mathbf{x})$:

- Aggregate via averaging (regression) or majority voting (classification)
- Posterior class probabilities $\hat{\pi}_k(\mathbf{x})$ can be estimated by calculating predicted class frequencies over the ensemble



WHY/WHEN DOES BAGGING HELP?

In one sentence:

Because the variability of the average of the predictions of many base learner models is smaller than the variability of the predictions from one such base learner model.

If the error of a base learner model is mostly due to (random) variability and not due to structural reasons, combining many such base learners by bagging helps reducing this variability.

Bagging algorithm: Training

Input: Dataset $\mathcal{D}$, type of BLs, number of bootstraps M

for $m = 1, \ldots, M$ **do**

  Draw a bootstrap sample $\mathcal{D}^{[m]}$ from $\mathcal{D}$

  Train a base learner $\hat{f}^{[m]}$ on $\mathcal{D}^{[m]}$

end for

Bagging algorithm: Prediction

Input: Obs. $\mathbf{x}$, trained BLs $\hat{b}^{[m]}$ (as scores $\hat{f}^{[m]}$, hard labels $\hat{h}^{[m]}$ or probs $\hat{\pi}^{[m]}$)

$$\hat{f}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{f}^{[m]}(\mathbf{x}) \right) \qquad \text{(regression / decision score, use } \hat{f}_k \text{ in multi-class)}$$

$$\hat{h}(\mathbf{x}) = \arg\max_{k \in \mathcal{Y}} \sum_{m=1}^{M} \mathbb{I}\left(\hat{h}^{[m]}(\mathbf{x}) = k \right) \qquad \text{(majority voting)}$$

$$\hat{\pi}_k(\mathbf{x}) = \begin{cases} \frac{1}{M} \sum_{m=1}^{M} \hat{\pi}_k^{[m]}(\mathbf{x}) & \text{(probabilities through averaging)} \\ \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}\left(\hat{h}^{[m]}(\mathbf{x}) = k \right) & \text{(probabilities through class frequencies)} \end{cases}$$

WHY/WHEN DOES BAGGING HELP? / 2

- Bagging reduces the variability of predictions by averaging the outcomes from multiple BL models
- It is particularly effective when the errors of a BL are mainly due to (random) variability rather than systematic issues

Assume we use quadratic loss and measure instability of the ensemble with

$$\Delta\left(f^{[M]}(\mathbf{x})\right) = \frac{1}{M}\sum_{m}^{M}\left(b^{[m]} - f^{[M]}(\mathbf{x})\right)^2$$

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$$= \frac{1}{M}\sum_{m}^{M}\left(\left(b^{[m]} - y\right) + \left(y - f^{[M]}(\mathbf{x})\right)\right)^2$$

$$= \frac{1}{M}\sum_{m}^{M} L(y, b^{[m]}) + L(y, f^{[M]}(\mathbf{x})) \underbrace{-2\left(y - \frac{1}{M}\sum_{m=1}^{M} b^{[m]}\right)\left(y - f^{[M]}(\mathbf{x})\right)}_{-2L(y, f^{[M]}(\mathbf{x}))}$$

So, if we take the expected value over the data's distribution:

$$\mathrm{E}_{xy}\left[L\left(y, f^{[M]}(\mathbf{x})\right)\right] = \frac{1}{M}\sum_{m}^{M}\mathrm{E}_{xy}\left[L\left(y, b^{[m]}\right)\right] - \mathrm{E}_{xy}\left[\Delta\left(f^{[M]}(\mathbf{x})\right)\right]$$

$\Rightarrow$ The expected loss of the ensemble is lower than the average loss of the single base learner by the amount of instability in the ensemble's base learners.

The more accurate and diverse the base learners, the better.

- Increasing # of BLs improves performance until a point
- Optimal ensemble size depends on inducer and data distribution

IMPROVING BAGGING

Bagged ensembles with 100 BLs each on spam:

Bagging seems especially helpful for less stable learners like CART

How to make $\mathrm{E}_{xy}\left[\Delta\left(f^{[M]}(\mathbf{x})\right)\right]$ as large as possible?

$$\mathrm{E}_{xy}\left[L\left(y, f^{[M]}(\mathbf{x})\right)\right] = \frac{1}{M}\sum_{m}^{M}\mathrm{E}_{xy}\left[L\left(y, b^{[m]}\right)\right] - \mathrm{E}_{xy}\left[\Delta\left(f^{[M]}(\mathbf{x})\right)\right]$$

Assume $\mathrm{E}_{xy}\left[b^{[m]}\right] = 0$ for simplicity, $\mathrm{Var}_{xy}\left[b^{[m]}\right] = \mathrm{E}_{xy}\left[(b^{[m]})^2\right] = \sigma^2$, $\mathrm{Corr}_{xy}\left[b^{[m]}, b^{[m]}m'\right] = \rho$ for all m, m'.

$$\Longrightarrow \mathrm{Var}_{xy}\left[f^{[M]}(\mathbf{x})\right] = \frac{1}{M}\sigma^2 + \frac{M-1}{M}\rho\sigma^2 \qquad \left(\ldots = \mathrm{E}_{xy}\left[(f^{[M]}(\mathbf{x}))^2\right]\right)$$

$$\begin{aligned}
\mathrm{E}_{xy}\left[\Delta\left(f^{[M]}(\mathbf{x})\right)\right] &= \frac{1}{M}\sum_{m}^{M}\mathrm{E}_{xy}\left[\left(b^{[m]} - f^{[M]}(\mathbf{x})\right)^2\right] \\
&= \frac{1}{M}\left(M\mathrm{E}_{xy}\left[(b^{[m]})^2\right] + M\mathrm{E}_{xy}\left[(f^{[M]}(\mathbf{x}))^2\right] - 2M\mathrm{E}_{xy}\left[b^{[m]}f^{[M]}(\mathbf{x})\right]\right) \\
&= \sigma^2 + \mathrm{E}_{xy}\left[(f^{[M]}(\mathbf{x}))^2\right] - 2\frac{1}{M}\sum_{m'}^{M}\underbrace{\mathrm{E}_{xy}\left[b^{[m]}b^{[m]}m'\right]}_{=\mathrm{Cov}_{xy}\left[b^{[m]}, b^{[m]}m'\right] + \mathbb{E}_{xy}\left[b^{[m]}\right]\mathbb{E}_{xy}\left[b^{[m]}m'\right]} \\
&= \sigma^2 + \left(\frac{1}{M}\sigma^2 + \frac{M-1}{M}\rho\sigma^2\right) - 2\left(\frac{M-1}{M}\rho\sigma^2 + \frac{1}{M}\sigma^2 + 0\cdot 0\right) \\
&= \frac{M-1}{M}\sigma^2(1-\rho)
\end{aligned}$$